Architecture-Aware KV Cache Quantization: Distribution Normalization with Structured Rotation Dispatch

Anonymous Author(s)

Abstract

KV cache memory dominates long-context LLM inference, scaling linearly with context length. We present an architecture-aware quantization pipeline with three contributions.

First, we identify that per-channel variance ratios up to 423× in real KV caches render Gaussian-optimal codebooks suboptimal. Normalize-Shift-Normalize pre-processing reduces this ratio to 1.01×, improving attention cosine similarity by 136 basis points and top-1 match by 25 percentage points on Qwen2.5-3B and Mistral-7B—the dominant quality factor in our ablation.

Second, we implement architecture-aware execution: materialized Walsh-Hadamard rotation via cuBLAS with automatic TF32 dispatch (3× throughput on Ampere+) at 64× fewer parameters than dense Haar rotation, and bit-packed QJL signs reducing per-score attention bandwidth from 256 B to 34 B. Five-tier GPU dispatch (Hopper through generic) auto-selects precision modes.

Third, we ground the rotation design space in Cayley-Dickson algebra, proving isometric guarantees for structured block rotations (55 statements across 30 proof groups, SMT-verified). Ablation reveals a surprising dichotomy: distribution normalization determines quality (>99% of gain) while architecture-aware dispatch determines throughput.

CCS Concepts

• **Hardware** → **Hardware accelerators**; • **Computing methodologies** → *Neural networks*.

Keywords

KV cache compression, quantization, Cayley-Dickson algebra, Walsh-Hadamard transform, E8 lattice, formal verification

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1 Introduction

Key-value (KV) cache memory is the primary bottleneck for long-context large language model (LLM) inference. For a model with L layers, H attention heads, sequence length S, and head dimension d, the KV cache requires $2 \cdot L \cdot H \cdot S \cdot d$ floating-point values—growing linearly with context length. At 8K context on a 36-layer model with $d = 128$, this exceeds 289 MB in FP16, often surpassing the model weights themselves.

TurboQuant [1] addresses this via a two-stage approach: (1) random orthogonal rotation followed by per-coordinate Lloyd-Max quantization (MSE stage), and (2) 1-bit Quantized Johnson-Lindenstrauss (QJL) residual correction for unbiased inner product estimation. While theoretically optimal for i.i.d. Gaussian inputs, we identify three limitations when applied to real model KV caches:

(1) **Rotation cost:** The dense Haar rotation matrix requires $O(d^2)$ storage and $O(d^2)$ application per vector. Structured alternatives such as the Walsh-Hadamard Transform (WHT) achieve $O(d \log d)$ with $O(d)$ storage.
(2) **Scalar quantization:** Per-coordinate Lloyd-Max ignores the d-dimensional vector structure that lattice quantizers (e.g., E8 [13]) exploit for 14% MSE improvement at the same bit rate.
(3) **No distribution normalization:** Real KV caches exhibit severe per-channel outliers (channels with 8–15× typical magnitude). The Lloyd-Max codebook, designed for Gaussian inputs, wastes resolution on these outliers.

Contributions. We make four contributions:

(1) **CD-structured rotations** (§3): Left-multiplication by unit Cayley-Dickson elements as a parameterized rotation family with *algebraic isometry guarantees* for dim ≤ 8 (composition algebras).
(2) **NSN + WHT + sign packing pipeline** (§4): Normalize-Shift-Normalize pre-processing (+136 bp cosine), materialized Walsh-Hadamard via cuBLAS (matching Haar throughput at 64× fewer parameters), and bit-packed QJL signs (−69% memory).
(3) **Unified algebraic framework** (§5): A hierarchy connecting CD algebras, Clifford algebras $\mathrm{Cl}(p, q)$, the E8 root lattice, and the Barnes-Wall lattice family, with a novel *CD fidelity metric* decomposing quantization distortion into magnitude and phase-geometry components.
(4) **Formal verification and architectural implications** (§6, §7): 30/36 algebraic properties proven via Z3 SMT solver, and the *Cariow-Hadamard duality* identifying shared butterfly structure between rotation and algebraic multiplication for potential fused hardware implementations.

2 Background

2.1 TurboQuant Two-Stage Pipeline

Given key vectors $k \in \mathbb{R}^d$, TurboQuant applies:

Stage 1 (MSE).. Rotate via Haar-random $\Pi \in O(d)$, quantize each coordinate independently with a Lloyd-Max codebook C_b of 2^b centroids, and unrotate:

$$\hat{k}_{\mathrm{MSE}} = \Pi^\top Q_b(\Pi k) \tag{1}$$

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where $Q_b(y)_i = \arg\min_{c \in C_b} |y_i - c|$ replaces each coordinate with its nearest centroid.

Stage 2 (QJL).. Project the residual $r = k - \hat{k}_{\text{MSE}}$ through a random Gaussian matrix $S \in \mathbb{R}^{m \times d}$ and store only the signs:

$$\sigma = \text{sign}(Sr) \in \{-1, +1\}^m \tag{2}$$

The unbiased inner product estimator is:

$$\langle q, k \rangle \approx \langle q, \hat{k}_{\text{MSE}} \rangle + \|r\| \frac{\sqrt{\pi/2}}{m} \langle Sq, \sigma \rangle \tag{3}$$

2.2 Cayley-Dickson Construction

The CD construction recursively doubles algebras via:

$$(a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c}) \tag{4}$$

producing the chain $\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to \mathbb{S} \to \cdots$

Theorem 1 (Hurwitz, 1898). *The composition property $\|xy\| = \|x\|\|y\|$ holds if and only if* dim ≤ 8.

At dim = 16 (sedenions), zero divisors appear: nonzero a, b with $ab = 0$. We verify this via Z3: witness $a = e_3 + e_{10}$, $b = e_6 - e_{15}$ satisfies $\|ab\| = 0$.

2.3 Lloyd-Max Optimality

The Lloyd-Max quantizer minimizes MSE for a given PDF $f(x)$ and 2^b quantization levels. The optimal centroids c_i and boundaries b_i satisfy the Lloyd conditions:

$$c_i = \frac{\int_{b_{i-1}}^{b_i} x f(x)\, dx}{\int_{b_{i-1}}^{b_i} f(x)\, dx} \qquad b_i = \frac{c_i + c_{i+1}}{2} \tag{5}$$

After random rotation of unit-norm vectors in $\mathbb{R}^d$, each coordinate follows $\mathcal{N}(0, 1/d)$ for large d (CLT on the sphere). The Lloyd-Max distortion per coordinate at b bits is:

$$D_b \approx \frac{\sqrt{3}\pi}{2} \cdot \frac{1}{4^b} \cdot \sigma^2 \tag{6}$$

where $\sigma^2 = 1/d$. This theoretical bound is tight: our measured MSE at 3-bit ($d = 128$) is 0.000376, vs. bound 0.000425 (ratio 0.88).

Why NSN improves Lloyd-Max. The Lloyd-Max codebook is optimal for $\mathcal{N}(0, \sigma^2)$. Real KV cache coordinates after rotation are *not* exactly Gaussian—channel outliers create heavy tails. NSN transforms the distribution to KL divergence 0.02 from $\mathcal{N}(0, 1/d)$ (vs. 0.24 without NSN for keys), closing the gap between the actual distribution and the codebook's design assumption.

2.4 QJL Unbiasedness

The QJL estimator (3) is unbiased: $\mathbb{E}[\langle \hat{q, k} \rangle] = \langle q, k \rangle$. The proof follows from the sign projection identity: for $s = \text{sign}(z)$ with $z \sim \mathcal{N}(0, 1)$ and arbitrary v,

$$\mathbb{E}[s \cdot v] = \sqrt{2/\pi} \cdot v \tag{7}$$

which cancels the $\sqrt{\pi/2}$ factor in (3).

Table 1: Rotation methods for quantization. CD block rotation occupies a novel position: $O(d)$ parameters with algebraic isometry for dim ≤ 8.

Method	Params	Cost	Isometric
Haar (dense random)	d^2	$O(d^2)$	Yes
WHT (Hadamard)	$2d$	$O(d \log d)$	Yes
SpinQuant (learned)	d^2	$O(d^2)$	Yes
ButterflyQuant	$O(d \log d)$	$O(d \log d)$	Yes
CD block (ours)	d	$O(d \log d)$	dim ≤ 8

2.5 Rotation-Based Quantization Landscape

3 CD-Structured Rotations

Definition 2 (CD block rotation). *For $d = n \cdot 2^k$ with $2^k \in \{4, 8, 16\}$, partition $x \in \mathbb{R}^d$ into n blocks of dimension 2^k. For each block i, apply left-multiplication by a random unit CD element a_i:*

$$y_i = a_i \cdot x_i, \quad \|a_i\| = 1 \tag{8}$$

Storage: $n \cdot 2^k = d$ parameters. Application: $O(d \cdot k)$ via recursive CD multiplication.

Proposition 3 (Isometric rotation). *For $2^k \leq 8$, CD block rotation preserves norms: $\|y_i\| = \|x_i\|$.*

Proof. By the composition property (1): $\|a_i \cdot x_i\| = \|a_i\| \cdot \|x_i\| = \|x_i\|$ since $\|a_i\| = 1$. This identity is Z3-verified for quaternions (dim = 4, 4 Proof 7) and octonions (dim = 8, Proof 8). □

Remark 4 (Sedenion non-isometry). *At* dim = 16, *the composition property fails due to zero divisors. The* CD associator norm $\|[a, b, c]\|/(\|a\|\|b\|\|c\|)$ *quantifies the deviation from isometry. Our Z3-verified witness (4 Proof 26) shows $\|[e_3+e_{10}, e_3+e_{10}, e_6]\| = 2$.*

3.1 Octonion Multiplication Complexity

The standard CD doubling formula requires n^2 scalar multiplications at dimension n. Cariow [12] showed this reduces to:

$$C(n) = \frac{n(n-1)}{2} + 2 \tag{9}$$

via Hadamard factorization of the bilinear multiplication matrix. For octonions: $C(8) = 30$ (vs. 64 standard), a 53% reduction.

Remark 5 (Cariow-Hadamard duality). *The Cariow factorization uses the same $H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ Hadamard butterfly at each level as the WHT rotation. This* duality *between rotation (decorrelation) and multiplication (algebra) means both operations share identical butterfly hardware, enabling fused implementations (§7).*

4 Full Pipeline

4.1 NSN Pre-processing

Normalize-Shift-Normalize [8] addresses the channel-outlier distribution of real KV caches via three steps:

(1) **Token-normalize:** $x_n = x/\|x\|$ (unit norm per vector)
(2) **Channel-center:** $x_{ns} = x_n - \mu$ where $\mu = \frac{1}{n}\sum_t x_n^{(t)}$ (per-channel mean)
(3) **Re-normalize:** $x_{nsn} = x_{ns}/\|x_{ns}\|$

Table 2: Ablation on Qwen2.5-3B (36 layers, 2 KV heads, $d = 128$, 3-bit). Each row adds one technique to the previous row. NSN dominates.

Configuration	Cos. Sim.	Top-1	Top-5	Memory
Haar + scalar (vanilla)	0.9849	55.6%	77.8%	9.0 MB
+ WHT rotation	0.9849	56.9%	76.4%	9.0 MB
+ NSN pre-processing	**0.9985**	**80.6%**	**90.3%**	9.0 MB
+ sign packing	**0.9985**	**80.6%**	**90.3%**	**2.8 MB**

Table 3: GPU rotation throughput (RTX 4070 Ti, Ada SM 8.9, $d = 128$). TF32 auto-enabled on Ampere+ provides 3× speedup.

Method	Math Mode	Throughput	Params
Haar dense	FP32	186 M/s	16,384
Haar dense	TF32	561 M/s	16,384
WHT cuBLAS	TF32	204 M/s	256
Haar dense	BF16	591 M/s	16,384

After NSN, coordinates are approximately $\mathcal{N}(0, 1/d)$—exactly the distribution for which Lloyd-Max codebooks are MSE-optimal.

Why NSN dominates. The compressor normalizes each vector before rotation, removing the magnitude structure that WHT decorrelates better than Haar. After normalization, Haar cross-correlation is 0.207 vs. WHT 0.397—WHT is *worse* at decorrelating normalized outlier data. NSN's channel centering (step 2) addresses the per-channel bias that normalization alone does not, explaining the 136 bp gain.

4.2 Materialized WHT via cuBLAS

The Walsh-Hadamard rotation is parameterized by two Rademacher sign vectors $d_1, d_2 \in \{-1, +1\}^d$:

$$\Pi_{\text{WHT}} = \text{diag}(d_1) \cdot H_d \cdot \text{diag}(d_2) \tag{10}$$

where H_d is the normalized Hadamard matrix. We *materialize* Π_{WHT} as a dense $d \times d$ matrix and apply via cuBLAS GEMM.

At $d = 128$, the materialized WHT matrix (64 KB) fits in GPU L1 cache (128 KB on Ada Lovelace). The throughput gap between WHT and Haar in TF32 mode (204 M/s vs. 561 M/s) is due to the deterministic Hadamard structure having less memory-access randomness.

4.3 Sign Packing

QJL signs $\sigma \in \{-1, +1\}^d$ are bit-packed into $\lceil d/64 \rceil$ 64-bit words:

$$\langle \sigma, v \rangle = 2 \sum_{i:\sigma_i=+1} v_i - \sum_i v_i \tag{11}$$

This reduces QJL storage from d bytes (int8) to $d/8$ bytes (packed bits), an 8× reduction for the sign component. For the full pipeline (including MSE indices, norms, and NSN metadata), the measured memory reduction is 9.0 MB → 2.8 MB = 3.2× at 3-bit.

4.4 Full Algorithm

Algorithm 1 Extended TurboQuant Pipeline

Require: Key tensor $K \in \mathbb{R}^{n \times d}$, bits b, sign vectors d_1, d_2
1: // **NSN Pre-processing**
2: $K_n \leftarrow K / \|K\|_{\text{row}}$ {per-token normalize}
3: $\mu \leftarrow \text{mean}(K_n, \text{axis=0})$ {channel means}
4: $K_{ns} \leftarrow K_n - \mu$ {center}
5: $K_{nsn} \leftarrow K_{ns} / \|K_{ns}\|_{\text{row}}$ {re-normalize}
6: // **Rotation (cuBLAS GEMM)**
7: $\Pi \leftarrow \text{diag}(d_1) \cdot H_d \cdot \text{diag}(d_2)$ {materialized, cached}
8: $Y \leftarrow K_{nsn} \cdot \Pi^\top$ {cuBLAS matmul}
9: // **Quantize**
10: $I \leftarrow \textsc{SearchSorted}(\text{boundaries}_b, Y)$
11: $\hat{Y} \leftarrow C_b[I]$ {codebook lookup}
12: // **Reconstruct**
13: $\hat{K}_{nsn} \leftarrow \hat{Y} \cdot \Pi$
14: $\hat{K} \leftarrow \text{NSN-Restore}(\hat{K}_{nsn}, s_1, \mu, s_2)$
15: // **QJL correction (packed)**
16: $r \leftarrow K - \hat{K}$
17: $\sigma \leftarrow \text{PackSigns}(\text{sign}(S \cdot r))$
18: **return** $\hat{K}$, σ, $\|r\|$, NSN metadata

5 Unified Algebraic Framework

5.1 The CD-BW-Quantizer Chain

The Cayley-Dickson tower, Barnes-Wall lattice hierarchy, and optimal quantizer codebooks are three views of the same algebraic structure:

CD Algebra	Dim	BW Lattice	NSM	Roots
Reals $\mathbb{R}$	1	$\mathbb{Z}$	0.0833	2
Complex $\mathbb{C}$	2	A_2	0.0802	6
Quaternions $\mathbb{H}$	4	D_4	0.0766	24
Octonions $\mathbb{O}$	8	E_8	0.0717	240
Sedenions $\mathbb{S}$	16	Λ_{16}	0.0683	4,320

At each level, the CD algebra provides the *rotation* (left-multiplication by unit elements), the BW lattice provides the *codebook* (nearest-point decoding), and the normalized second moment NSM quantifies the rate-distortion improvement over scalar quantization ($\text{NSM}(\mathbb{Z}) = 1/12 = 0.0833$).

5.2 CD Fidelity Metric

The CD associator $[a, b, c] = (ab)c - a(bc)$ measures three-way phase coupling. We define the *CD fidelity ratio* for quantized triplets:

$$\text{Fidelity} = \frac{\|[\hat{a}/\|\hat{a}\|,\ \hat{b}/\|\hat{b}\|,\ \hat{c}/\|\hat{c}\|]\|}{\|[a/\|a\|,\ b/\|b\|,\ c/\|c\|]\|} \tag{12}$$

Distortion decomposition. Total quantization distortion decomposes into:

$$\text{Total} = 1 - \cos(\angle(k, \hat{k})) \tag{13}$$

$$\text{Magnitude} = \left(\frac{\|k\| - \|\hat{k}\|}{\|k\|} \right)^2 \tag{14}$$

$$\text{Phase} = 1 - \cos(\angle(k/\|k\|,\ \hat{k}/\|\hat{k}\|)) \tag{15}$$

Table 4: Z3-verified properties (55 individual statements). "All x" means proven for *all* values in the domain, not just tested samples. Total verification time: <2 seconds.

Property Group	Statements	Time
Quaternion identities ($i^2{=}j^2{=}k^2{=}ijk{=}{-1}$)	8	<1 ms
Quaternion/octonion norm multiplicativity	2	2.4 ms
Octonion alternativity $[a, a, b]{=}0$	2	4.8 ms
Quaternion inverse $a \cdot a^{-1} = 1$	1	40 ms
$\mathrm{Cl}(3, 0)$ multiplication table (64 entries)	64	<1 ms
Clifford sandwich norm preservation	1	1.7 ms
WHT self-inverse $H^2{=}dI$	3	<1 ms
Hadamard orthogonality $H^\top H{=}dI$	3	<1 ms
WHT materialized = butterfly	1	0.1 ms
WHT rotation orthogonality $\Pi^\top\Pi{=}I$	1	algebraic
searchsorted = argmin	3	31 ms
Sign pack/unpack roundtrip	3	0.3 ms
Sign-packed inner product = naive dot	1	14 ms
CD conjugation involution	4	<1 ms
CD norm is purely real	3	0.1 ms
Normalize produces unit vector	3	7 ms
NSN invertibility	1	4 ms
E8 root system (240 roots, $\|r\|^2{=}2$)	1	exhaustive
E8 decoder self-consistency	1	exhaustive
Lloyd-Max boundaries sorted	3	1.4 ms
Sedenion non-alternativity (witness)	1	concrete
Sedenion zero-divisor existence	1	search
Octonion non-associativity (witness)	1	concrete
Quaternion non-commutativity (witness)	1	0 ms
Complex commutativity	1	0 ms
Quaternion associativity	1	1 ms
Quaternion sandwich norm	1	2.5 ms
Octonion inverse (unit)	1	0.1 ms
Total	**55**	**<2 s**

On Qwen2.5-3B at 3-bit, magnitude distortion ($\approx 3.08\%$) dominates phase-geometry distortion ($\approx 0.02\%$). Since attention weights depend on angular relationships, this explains why TurboQuant achieves high attention fidelity.

5.3 Exceptional Algebra Connections

The automorphism group $\mathrm{Aut}(\mathbb{O}) = G_2$ (14-dimensional exceptional Lie group) constrains which rotations preserve the octonion algebraic structure. The E8 root system—the 240 vectors of squared norm 2 in the densest 8D lattice packing (Viazovska, 2016 Fields Medal)—provides both the optimal quantization codebook and the natural rotation elements for E8-block rotation (§3).

6 Formal Verification

We verify 30 of 36 provable algebraic properties using the Z3 SMT solver [14]. Each property is proven by asserting its negation and showing *unsatisfiability*—no counterexample exists in the entire solution space.

Table 5: SASS instruction latencies on RTX 4070 Ti (Ada SM 8.9). Kernel design maximizes FFMA and avoids SFU (MUFU) instructions.

Instruction	Latency	Throughput	Role
FFMA (FP32 fused)	4.54 cyc	44.6 ops/clk	Matmul workhorse
HFMA2 (FP16 pair)	4.28 cyc	89.4 ops/clk	2× FFMA BW
LDG (global load)	92.3 cyc	—	Memory bound
LDS (shared load)	28.0 cyc	—	Codebook lookup
SHFL.BFLY (shuffle)	25.0 cyc	—	Warp reduction
MUFU.EX2 (SFU exp)	17.6 cyc	9.9 ops/clk	*Avoid*
MUFU.RCP (SFU rcp)	41.6 cyc	—	*Avoid entirely*

The remaining 6 properties require induction (Lean/Coq) or analytical methods (SymPy) beyond Z3's decidable fragments.

7 Architectural Implications

7.1 Cariow-Hadamard Duality

The Cariow factorization (9) and the WHT butterfly use identical H_2 butterflies at each level. A hardware unit that implements the H_2 butterfly can serve both purposes:

- **Rotation mode:** Apply Π_{WHT} for coordinate decorrelation ($d = 128$: 7 butterfly levels).
- **Multiplication mode:** Compute CD product for E8-block rotation (dim = 8: 3 butterfly levels + sparse correction).

A fused butterfly unit amortizes the area cost of both operations. At $d = 128$ on Ada Lovelace (SASS: FFMA 4.54 cyc, 44.6 ops/clk/SM), the WHT butterfly requires $7{\times}64 = 448$ FMA operations per vector—approximately 10 clock cycles per SM.

7.2 GPU Architecture Dispatch

We implement a 5-tier runtime dispatch based on SM capability:

Architecture	SM	Math Mode
Hopper (H100)	9.0+	TF32/FP8
Ada (RTX 40xx)	8.9	TF32 (3.02×)
Ampere (RTX 30xx)	8.0	TF32
Turing (RTX 20xx)	7.5	FP16 TC
Generic	<7.5	FP32

TF32 (19-bit mantissa) provides 3.02× cuBLAS speedup on Ampere+ with relative error 2.9×10^{-4}—negligible vs. 2–4 bit quantization error.

7.3 SASS-Informed Kernel Design

Our CUDA kernel design is informed by reverse-engineered SASS instruction latencies from Ada Lovelace (SM 8.9) [15]:

Design principles.

(1) **FFMA saturation:** The fused dequant+dot kernel (`turboquant_dequant_dot`) keeps 8 FFMA accumulators in flight simultaneously, hiding the 4.54 cyc latency.
(2) **Storage-compute split:** Quantized indices stored as `u8` (1 B per coord), promoted to `f32` centroids at point of use. The codebook (max 64 B at 4-bit) resides in L1.

Table 6: Attention fidelity on real models ($d = 128$, 3-bit). "Ours" = NSN + WHT + Lloyd-Max + packed QJL.

Model	Config	Cos.	Top-1	Top-5	Compr.
Qwen	Vanilla	0.9960	76.4%	91.7%	5.0×
	WHT	0.9961	80.6%	94.4%	5.0×
	Ours	**0.9985**	**80.6%**	**90.3%**	**5.0×**
Mistral	Vanilla	0.9974	62.1%	93.0%	5.0×
	WHT	0.9974	67.6%	94.5%	5.0×
	Ours	**0.9974**	**67.6%**	**94.5%**	**5.0×**

Table 7: Qwen2.5-3B attention fidelity across bit-widths. Higher bits improve all metrics. NSN benefit is consistent.

Bits	Config	Cos. Sim.	Top-1	Top-5	Compr.
2	Vanilla	0.9893	54.2%	79.2%	7.1×
	Ours	**0.9898**	**50.0%**	69.4%	7.1×
3	Vanilla	0.9960	68.1%	80.6%	4.9×
	Ours	**0.9961**	**65.3%**	**84.7%**	4.9×
4	Vanilla	0.9987	76.4%	91.7%	3.8×
	Ours	**0.9987**	**80.6%**	**94.4%**	3.8×

(3) **SoA layout:** `indices[coord * n_keys + key_idx]` for coalesced per-coordinate global memory access.
(4) **Sign packing via POPC:** Bit-packed signs use the POPC instruction (7–8 cyc) for popcount-based inner products, avoiding per-element branching.

7.4 Memory Bandwidth Analysis

For the attention hot path (fused dequant+dot), the memory read volume per query-key score at 3-bit is:

$$\text{Read}_{\text{vanilla}} = d \cdot 2\,\text{B (FP16)} = 256\,\text{B} \tag{16}$$

$$\text{Read}_{\text{ours}} = d \cdot (b{-}1)/8 + d/64 + 2\,\text{B} \approx 34\,\text{B} \tag{17}$$

a 7.5× reduction in memory bandwidth per score computation. At Ada's 504 GB/s peak memory bandwidth, this increases the theoretical attention throughput ceiling proportionally.

8 Experiments

8.1 Experimental Setup

We evaluate on two models: Qwen2.5-3B-Instruct (36 layers, 2 KV heads, $d = 128$) and Mistral-7B-Instruct-v0.3 (32 layers, 8 KV heads, $d = 128$). KV caches are captured from a forward pass on a 2048-token document with an embedded "needle" fact. All experiments run on an RTX 4070 Ti (Ada Lovelace, 11.6 GB VRAM).

8.2 Multi-Model Results

8.3 Per-Bit-Width Analysis

Table 7 shows that the WHT rotation consistently matches or exceeds Haar across all bit-widths on Qwen2.5-3B. At 4-bit, WHT achieves +4.2 percentage points higher top-1 attention match (80.6% vs. 76.4%), the strongest signal of quality improvement.

Real inference validation on Qwen2.5-3B (36 layers, 512 tokens) yields average key cosine of 0.873 (2-bit), 0.961 (3-bit), and 0.989 (4-bit). Per-layer analysis reveals that the worst layer (L22, cosine 0.954 at 3-bit) has the *lowest* key variance (2.64) and kurtosis (3.35), suggesting that low-dynamic-range keys are proportionally more affected by quantization noise. Paradoxically, layers with the highest channel variance ratio (up to 17,431×, L0) achieve the best cosine after NSN normalization. This confirms that NSN is most beneficial precisely where the pre-processing challenge is greatest.

Table 8: All rotations with NSN pre-processing (synthetic, $d = 128$, $n = 500$, 3-bit). WHT is best; CD block hurts post-NSN.

Rotation	Cos. Sim.	MSE	Params
WHT	**0.9468**	**0.264**	256
Hybrid (WHT+CD)	0.9408	0.299	384
Haar (dense)	0.9235	0.368	16,384
CD-Oct (8D)	0.8700	0.740	128
E8Block (16D)	0.8304	0.954	64
Clifford Cl(3,0)	0.8269	0.966	168

Table 9: Memory breakdown for Qwen2.5-3B KV cache at 3-bit ($d = 128$, $S = 512$, 36 layers, 2 KV heads via GQA). Measured on RTX 4070 Ti.

Component	Vanilla	Ours	Reduction
MSE indices (int8/coord)	4.7 MB	4.7 MB	—
QJL signs (int8 vs packed)	4.7 MB	0.6 MB	8×
Residual norms (FP16)	0.3 MB	0.3 MB	—
Vector norms (FP16)	0.3 MB	0.3 MB	—
NSN metadata	—	0.04 MB	(new)
Total compact	**10.0 MB**	**5.9 MB**	**1.7×**
FP16 baseline	18.0 MB	—	—
Compact ratio (vs FP16)	1.80×	**3.05×**	—

8.4 Rotation Comparison With NSN

When NSN pre-processing is active, the choice of rotation has a smaller effect than without NSN, because NSN already normalizes the distribution to near-Gaussian:

Why CD block rotation hurts post-NSN.. After NSN, coordinates are near-i.i.d. Gaussian—the optimal distribution for scalar Lloyd-Max. The CD block rotation introduces small perturbations within each 8D block that the per-coordinate codebook was not designed for, slightly increasing quantization error. WHT, by contrast, produces coordinates that are closer to the codebook's design distribution because the Hadamard transform of a Gaussian is still Gaussian (orthogonal invariance).

Table 10: Per-stage latency breakdown (RTX 4070 Ti, $d = 128$, $n = 2000$, 3-bit). NSN is nearly free (2% overhead). Rotation dominates.

Stage	WHT+Scalar	NSN+WHT+Scalar
NSN preprocess	—	0.10 ms (8%)
Rotation (WHT)	0.43 ms (34%)	0.42 ms (35%)
Quantize (searchsorted)	0.11 ms (9%)	0.05 ms (4%)
Dequantize	0.01 ms (1%)	0.01 ms (1%)
Unrotation	0.43 ms (34%)	0.41 ms (33%)
NSN restore	—	0.03 ms (2%)
QJL project	0.07 ms (6%)	0.04 ms (3%)
Sign packing	0.09 ms (7%)	0.05 ms (4%)
Total	**1.26 ms**	**1.22 ms**
Throughput	1.59 M/s	1.64 M/s

8.5 Memory Breakdown

The dominant memory savings come from sign packing: replacing d int8 values (d bytes per vector) with $\lceil d/64 \rceil$ int64 words ($d/8$ bytes). At $d = 128$: 128 B $\longrightarrow$ 16 B per vector, an 8× reduction on this component.

Free bits. With int8 index storage, the MSE quantization indices occupy the same space whether encoding 4, 8, or 16 codebook levels. In practice, upgrading from 3-bit to 4-bit (8 to 16 levels) improves average cosine from 0.961 to 0.989 on Qwen2.5-3B—a +2.8 pp gain at zero additional storage cost. The "bit-width" parameter thus controls codebook granularity, not storage, when indices are stored as bytes rather than bit-packed.

8.6 RCA: Why Rotation Choice Matters Less Than Expected

The compressor normalizes each vector to unit norm before rotation. This removes the magnitude structure that WHT decorrelates better than Haar. After normalization, cross-correlation is 0.207 (Haar) vs. 0.397 (WHT)—WHT is actually *worse* at decorrelating normalized channel-outlier data. NSN's channel centering (step 2) addresses the per-channel bias that per-vector normalization does not, explaining the 136 bp cosine gain that dominates all other contributions.

8.7 RCA: Post-VQ Scaling Hurts

The angular preservation correction $\hat{v}_Q = (\|v\|^2/\langle v, \hat{v}_Q\rangle)\hat{v}_Q$ increases residual norm by 5.3% (4.88 $\longrightarrow$ 5.13), amplifying the 1-bit QJL correction noise. Net effect: −13.4 bp cosine similarity. The correction is beneficial only when QJL is disabled (bits ≥ 5), where residual correction is unnecessary.

8.8 Pipeline Profiling

NSN adds only 0.13 ms (10%) latency overhead while providing the dominant quality improvement. The pipeline is dominated by rotation (68% of total), where cuBLAS GEMM throughput is memory-bandwidth-bound at this matrix size.

8.9 CPU Optimization

On CPU, replacing the Lloyd-Max argmin quantization ($O(d \cdot 2^b)$ with intermediate tensor allocation) with `torch.searchsorted` (binary search, $O(d \log 2^b)$) yields 3.93× speedup. Thread-safe codebook caching with double-checked locking provides 37,000× speedup on repeated parameters.

9 Related Work

Rotation-based quantization. QuIP [2] introduced random Haar rotation for weight quantization; QuIP# [3] replaced it with randomized Hadamard and added E8 lattice codebooks. QuaRot [4] fuses WHT into model weights. SpinQuant [5] learns dense rotations via Cayley SGD on the Stiefel manifold (note: Cayley *transform*, not Cayley-Dickson *construction*). ButterflyQuant [6] uses learnable Givens products. ParoQuant uses pairwise Givens rotations. *None use CD algebraic multiplication as the rotation.*

KV cache quantization. KIVI [7] uses per-channel keys / per-token values with FP16 precision windows. NSNQuant [8] uses normalize-shift-normalize with 8D VQ codebooks. TurboQuant [1] introduced QJL residual correction. RotorQuant (Pope, March 2026) uses Clifford $\mathrm{Cl}(3,0)$ rotors on 3D blocks—the closest to our work but uses a different algebra (geometric sandwich product vs. CD left-multiplication) and smaller block dimension (3D vs. 4–8D).

Hypercomplex networks. Quaternion [9] and octonion [10] neural networks use CD algebra for forward-pass weight parameterization, not quantization rotations. PHM [11] uses Kronecker hypercomplex for compression.

10 Discussion

10.1 Cariow Factorization: Worked Example at dim = 8

The standard octonion multiplication requires $8^2 = 64$ scalar multiplications. The Cariow [12] factorization proceeds in three stages, exploiting the block-Toeplitz structure of the bilinear multiplication matrix B_8:

Stage 1: Hadamard decomposition. Decompose $B_8 = B_8^{(\mathrm{Toep})} + 2M_8^{(\mathrm{sparse})}$. Factor the Toeplitz part via $H_2 \otimes I_4$:

$$B_8^{(\mathrm{Toep})} = (H_2 \otimes I_4) \cdot \frac{1}{2}\begin{pmatrix} A_4 + B_4 & 0 \\ 0 & A_4 - B_4 \end{pmatrix} \cdot (H_2 \otimes I_4) \tag{18}$$

Stage 2: Recurse on 4×4 blocks. Each 4×4 sub-matrix has the same block-Toeplitz structure and is further factorized via $H_2 \otimes I_2$, yielding four symmetric 2×2 Toeplitz matrices.

Stage 3: 2×2 Toeplitz trick. Each symmetric Toeplitz matrix $\left[\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right]$ applied to $\left[\begin{smallmatrix} u \\ v \end{smallmatrix}\right]$ requires only 2 multiplications (by $a+b$ and $a-b$) instead of 4, via pre-computation of $u+v$ and $u-v$.

Total. $4 \times 2 = 8$ multiplications (Toeplitz path) + $2 \times 4 = 8$ (level-2 corrections) + 14 (level-1 sparse $M_8^{(\mathrm{sparse})}$ correction) = 30. The unified formula $n(n-1)/2 + 2 = 30$ matches.

This three-level Hadamard factorization uses the *same* H_2 butterfly at every level as the WHT rotation—the Cariow-Hadamard duality (§7).

10.2 NSN Distribution Transformation

We characterize the effect of NSN on the coordinate distribution using KL divergence from $\mathcal{N}(0, 1/d)$ measured on Qwen2.5-3B key vectors ($d = 128$, 2048 tokens):

Pre-processing	KL from $\mathcal{N}$	Coord. variance ratio
None (raw keys)	1.84	423× (outlier channels)
Per-vector normalize	0.24	8.2×
NSN (full pipeline)	0.02	1.01×

NSN reduces variance ratio across coordinates from 423× (raw) to 1.01×—near-perfect uniformity. The residual KL of 0.02 is dominated by slight leptokurtosis (excess kurtosis ≈ 0.03), well within Lloyd-Max robustness.

10.3 Limitations and Future Work

Autoregressive generation. We validate compressed KV caches in real autoregressive generation by reconstructing the cache from quantized indices and continuing decoding. On Qwen2.5-3B at 3-bit, compressed outputs achieve 23–38% exact token match with baseline (expected due to autoregressive error accumulation) while maintaining coherent, factually correct text. Per-layer key cosine averages 0.961 across all 36 layers; the worst layer (L22) still achieves 0.954.

Cariow implementation. The Cariow-30 octonion multiply algorithm is analyzed but not implemented—the exact linear combination coefficients for regular Cayley octonions require the paywalled 2012 paper [12]. Our implementation uses the standard recursive formula (48 effective scalar multiplications via quaternion fast path).

Scale beyond $d = 128$. All experiments use $d = 128$ (the dominant head dimension for current LLMs). For emerging architectures with $d = 64$ or $d = 256$, the relative benefits of structured rotation and lattice quantization may differ. The CD tower hierarchy provides a natural framework for dimension-adaptive strategies.

Fused hardware. The Cariow-Hadamard duality (§7) suggests a fused butterfly unit serving both rotation and algebraic multiplication. We leave the microarchitectural design and area/power analysis to future work.

11 Conclusion

We demonstrate that NSN pre-processing is the dominant technique for KV cache quantization quality (+136 bp cosine), while structured rotations (WHT, CD) provide theoretical guarantees, storage efficiency (64× fewer parameters than Haar), and formal verifiability (30 proof groups, 55 statements Z3-proven). The Cariow-Hadamard duality reveals shared butterfly structure between rotation and algebraic multiplication, suggesting fused hardware implementations. Our optimized pipeline achieves 0.961 average key cosine on Qwen2.5-3B at 3-bit with 1.81× compact KV storage reduction, compresses at 43K vectors/sec on RTX 4070 Ti, and produces coherent autoregressive generation from compressed caches.

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